Speech transmission over digital mobile radio channels

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The design of a speech channel for digital mobile radio applications is a trade-off between the key performance dimensions of speech quality, robustness to errors, delay, complexity and bit rate. An appropriate balance is often difficult to achieve, but is vital to customer satisfaction. This paper identifies the considerations in selecting a speech codec for mobile telephony applications, outlines techniques for robust and efficient speech transmission over a digital mobile radio channel and discusses how the resulting performance can be assessed. Throughout the paper, the half-rate GSM digital mobile radio system is used as an example.

1. Introduction

The last few years have witnessed a remarkable growth in the mobile telephony market, and digital systems now support a significant portion of the customer base. With the apparent ubiquity of digital technology, its application to mobile radio might be taken for granted. In practice, speech transmission over a digital mobile radio channel requires a difficult trade-off between a number of competing factors. This paper identifies the considerations in selecting a speech codec for mobile telephony applications, outlines techniques for robust and efficient speech transmission over a digital mobile radio channel and discusses how the resulting performance can be assessed.

There are currently 4.1 million mobile telephone customers in the United Kingdom (UK). Over 16% of these use one of the two digital systems, both of which are based on the European global system for mobile communications (GSM) standard. By the end of the century there will be an estimated 25 million GSM customers in western Europe alone, and GSM is also being adopted in eastern Europe, the Far East, Australia and Africa. The two GSM systems, GSM-900 and DCS-1800, are not available in North America, although licences have been granted for the PCS-1900 system, which uses GSM technology. (The suffixes in GSM-900, DCS-1800 and PCS-1900 refer to the system operating frequency in MHz.) GSM is certainly not the only digital mobile radio standard and a survey is provided in Wong et al [1]. However, it is currently the most widely adopted digital standard, and the recently defined half-rate speech channel will be used as an illustrative example throughout the paper.

Figure 1 highlights two of the key components for speech transmission over a digital mobile radio channel -- a speech codec and a channel codec. The function of the speech codec is to represent the speech signal using as few bits as possible. The performance of the speech codec determines an upper limit for the speech quality. However, poor radio reception introduces errors and the channel codec further encodes the speech data so that these errors can be detected and corrected. The degradation in speech quality under different error conditions is largely determined by the channel codec.

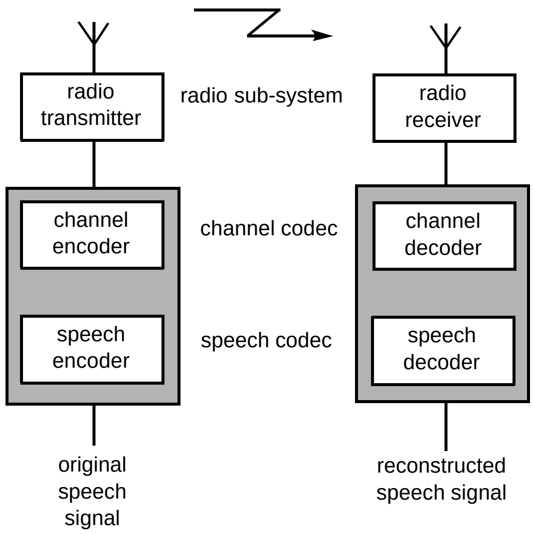

Fig 1 Key components for speech transmission over a digital mobile radio channel.

Section 2 of this paper discusses the performance trade-offs that must be made in a practical digital mobile radio system. Next, section 3 considers the requirements of a speech codec for mobile radio applications, while techniques for robust and efficient transmission are introduced in sections 4 and 5. As an illustrative example, section 6 describes some aspects of the half-rate GSM speech channel. In section 7 discussion covers how the performance of such a system can be measured. Finally, sections 8 and 9 consider future directions and conclusions.

2. Performance dimensions

The purpose of a communications system is to enable two or more people to communicate. This may sound obvious, but what is the definition of communication? In a military radio application, the requirement may simply be the reliable transfer of spoken information; radio operators can be trained in the use of the system and intelligibility is more important than naturalness. By contrast, in a telephony application, the customer requires a system that is simple to use and offers natural communication, conveying as many nuances of the original speech as possible. Consequently various dimensions of system performance must be balanced to suit the needs of the application in question. The most important performance dimensions of a digital speech transmission system are speech quality, robustness to errors, delay, complexity and bit rate [2].

The better the speech quality of a communications system, the more natural it will be to use. The distribution of available bits between the speech codec and channel codec depends very much on the application. In military radio, the ability to communicate at all times is paramount, while basic speech quality is of secondary importance. In telephony, good signal quality under normal operating conditions may be more important than providing service under extremely difficult radio conditions. The ratio of channel codec to speech codec bit rate is therefore likely to be higher in the military case.

Delay is also an important factor in determining the naturalness of a communications system. From a customer's point of view, the important figure is the delay between the production of a sound by the talker and the arrival of that sound at the ear of the remote listener. Delays that exceed a few hundred milliseconds (ms) rapidly lead to difficult and unnatural conversation. In cases of extreme delay, conversation may break down altogether. The ITU recommendations on delay are defined in Recommendation G.114.

Codec complexity can have a large impact on mobile terminal design. In digital mobile terminals, the power consumption of the radio sub-system is often small compared to that of the signal processing devices. A more complex codec may therefore reduce talk time or necessitate larger and heavier batteries. The number of silicon devices required to implement a codec can also affect the physical dimensions of the terminal. Ideally, it should be possible to implement a speech or channel codec on a single device. However, even in mass-production quantities, state-of-the-art high-speed devices can be extremely expensive.

In the development of speech codecs for the digital public switched telephone network (PSTN), the emphasis has been on maintaining good speech quality and low delay. The need for a cost-effective network has also meant low complexity codecs, with the result that the majority of speech traffic is currently carried at 64 kbit/s, although lower rates are sometimes used [1].

The data rate available to a user in a mobile radio system depends on a large number of factors including:

- the radio bandwidth allocated to the service, which is often determined by government or international agreement,
- the type of system, for example, cellular radio or satellite,
- the operating environment and associated radio propagation problems, such as multipath and shadowing,
- sources of radio interference, for example, other users of the system,
- limitations on radio transmission power, which could be regulatory or technical.

These and many other factors spanning physics, engineering and economics, combine to severely constrain the bit rate available to individual users in a practical radio system.

Obviously there are limits to which any one performance factor can be compromised. However, the restricted data rate and high occurrence of errors found in many mobile radio systems often dictate lower speech quality, greater complexity and longer delay than a PSTN link.

3. Speech codec considerations

A survey of speech codecs used in mobile radio systems is provided in Wong et al [1]. This section will consider the basic requirements of a speech codec suitable for use in a digital mobile radio system.

Codecs suitable for general audio often exploit the limitations of the human auditory system to provide near-transparent compression of arbitrary signals. Speech codecs achieve further compression by making judicious assumptions about the source signal to be encoded. A source model commonly used in speech codecs is the linear predictive model of speech production [3]. While the use of a source model reduces the bit rate of a codec, it also reduces the range of signals that can be faithfully transmitted. Problems arise when a speech codec is required to transmit signals that are not well described by the source model, and, as a consequence, transmission quality may be significantly degraded. In general, the lower the bit rate of a speech codec, the more dependent it will be on the assumptions of the source model.

3.1 Talker and language dependency

The use of source models can lead to codec performance that varies with different talkers and languages. The character of sounds produced by an individual talker are determined largely by their physiology. The parts of the human body that contribute to speech production vary substantially between individuals; significant differences are also observable as a function of gender and age. The physical mechanisms used to generate speech sounds can also differ between languages -- as do the nuances that carry the meaning. Ideally, a speech codec should be capable of faithfully encoding all possible human utterances, since the potential customer base of a public communications system spans the planet. In practice, design considerations often result in speech codec performance that is biased towards a particular language, gender or age group.

3.2 Signals other than speech

The typical acoustic environment of a mobile radio user differs significantly from that of a PSTN user. The sound pressure level (SPL) of ambient noise in an office or home environment is generally in the range of 35 to 55 dB(A). However, the SPL experienced by a mobile radio user, for example in a moving vehicle, aircraft, or in a busy street, may exceed 85 dB(A).

A high ambient noise level creates two problems. Firstly, it increases the listening effort required by the mobile radio user, since the signal produced by the handset ear piece must compete with the ambient noise. Secondly, it introduces noise into the speech received by the handset microphone. A speech codec selected for mobile radio applications should therefore exhibit acceptable performance for signals containing high acoustic noise levels. It should be noted that PSTN speech codecs may also have to carry mobile radio traffic and should also be capable of handling ambient noise.

A partial solution to the problem of ambient acoustic noise is to simply reduce the amount entering the codec. Careful acoustic design of a handset can dramatically reduce its sensitivity to ambient noise compared to the user’s speech. Even better noise rejection figures are possible using noise cancelling microphones [4]. Figure 2 shows the noise rejection achievable with a well-designed noise-cancelling handset. While rejection is modest above 2 kHz, the degree achieved at low frequencies is very valuable. For example, much of the ambient acoustic energy in a moving vehicle is present below 200 Hz. An alternative approach is to use signal processing techniques to achieve noise suppression [5]. However, substantial reductions in noise level without perceptible distortion of the speech signal are difficult to achieve.

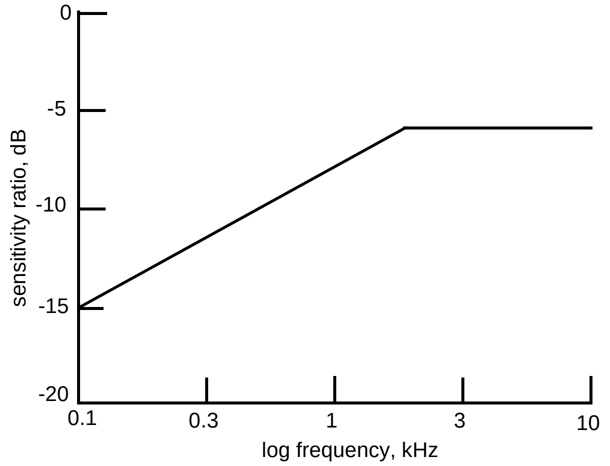

Fig 2 The ratio of te noise sesitivity to speech sensitivity for a well-designed noise-creating handset.

Further signals that may have to be transmitted include call progress tones (such as ring tone and busy tone), music on hold and dual tone multifrequency (DTMF) signals. The latter represent a particular problem since they must be reliably interpreted by a DTMF detection algorithm. In practice, many low bit rate speech codecs do not transmit DTMF tones reliably.

3.3 Codec tandeming issues

The term tandeming is used to describe the interconnection of multiple speech codecs. Tandeming can introduce substantial distortion into a link and is an important consideration in speech codec design.

Imagine the following scenario — a customer on an aircraft over the Pacific Ocean uses SkyphoneTM to call the UK. The call is routed to the UK through digital circuit multiplication equipment (DCME) and is answered by a voice mail service. Finally, the message is retrieved by a GSM customer. Figure 3 shows the speech codecs required to make such a call. As the number of mobile telephone users grows, scenarios such as this will occur with increasing frequency.

A problem for network planners is how to estimate the distortion introduced by a link such as the one in Fig 3. One method would be to measure the distortion introduced by each codec in isolation and sum the individual figures. This technique is sometimes used in PSTN circuits, where the performance of each codec is described in quantisation distortion units (qdu). Unfortunately, this approach is not valid for the type of distortion introduced by low bit rate

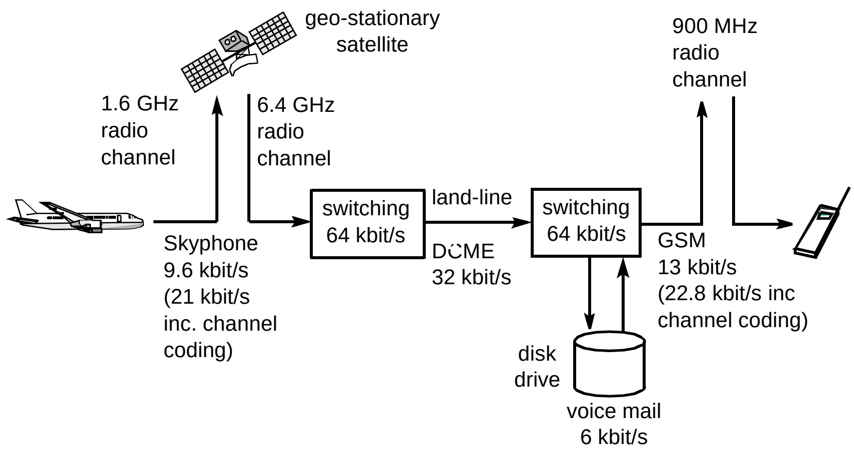

Fig 3 In this sceneario, the voice mail message left by the Skyphone customer passes through seven speech codecs before being heard by the GSM customer.

speech codecs, which cannot be treated as linear network elements [6, 7].

As mentioned earlier, many speech codecs rely on describing the input signal in terms of a source model. If the input to a codec is corrupted by the presence of ambient acoustic noise, or distortion introduced by previous codecs, the performance of the codec may be significantly worse than for clean speech. As a result, speech codecs in a tandem configuration may not perform as expected. In a digital mobile radio system connected to the PSTN, heterogeneous tandeming must be considered as well as homogeneous tandeming. The performance of a link containing heterogeneous speech codecs can even depend on the order in which they appear.

A second difficulty in estimating the subjective performance of speech codecs in tandem may occur if they exploit the properties of the human auditory system. Many speech codecs shape the frequency spectrum of their distortion so that it is rendered less audible. Unfortunately, each additional codec introduces further distortion, and there is a limit to how much distortion can be hidden from the listener using this technique. Consequently, a codec with very good subjective performance for a single coding may demonstrate significantly worse subjective performance after two or three codings.

4. Coping with channel errors

In this section the way in which the effects of channel errors are minimised in a digital mobile radio system are considered.

The bit error rate of a mobile radio channel can be substantially higher than would be tolerated in a PSTN channel. For example, in the PSTN bit error rates do not usually exceed 0.001%, whilst 0.1% is considered an exceptionally poor channel. However, in a radio system, the error rates experienced in areas of marginal reception can be as high as 10%. To compound the problem, moving terminals are subject to variable radio path loss. Periods of very low radio signal strength are referred to as fades and introduce error-bursts. Slow-moving terminals experience the longest fades, which may last for many hundreds of milliseconds.

4.1 Error correction and detection

To minimise the effect of channel errors, forward error correction (FEC) is applied to the speech data before it is transmitted. FEC increases the number of bits that must be transmitted but enables occasional errors to be corrected. Cyclic redundancy check (CRC) parity bits may also be added to the speech data prior to FEC to enable the detection of uncorrected errors.

Basic error protection can be provided by simple block coding. However, mobile radio channel codecs often employ convolutional encoding [8] combined with soft-decision Viterbi decoding [9]. These techniques are significantly more complex than block coding, but offer a much higher degree of error protection.

Not all of the bits that are generated by a speech encoder make an equal contribution to the reconstructed speech quality. This is particularly true for low bit rate speech codecs that represent the speech signal as a set of parameters. Therefore, to reduce the overhead of channel coding, FEC is often only applied to the speech bits that are subjectively most important.

4.2 Interleaving and frequency hopping

Although FEC is capable of correcting sparsely distributed bit errors, it cannot correct sustained periods of errors. Since the error characteristics of mobile radio channels are inherently bursty due to radio fades, the encoded data bits are spread across a number of transmission frames. This process is called interleaving and an example is shown in Fig 4. The number of transmission frames over which data is spread is referred to as the interleaving depth. However, interleaving introduces significant delay into the channel, and is generally only suitable for improving FEC performance over short radio fades.

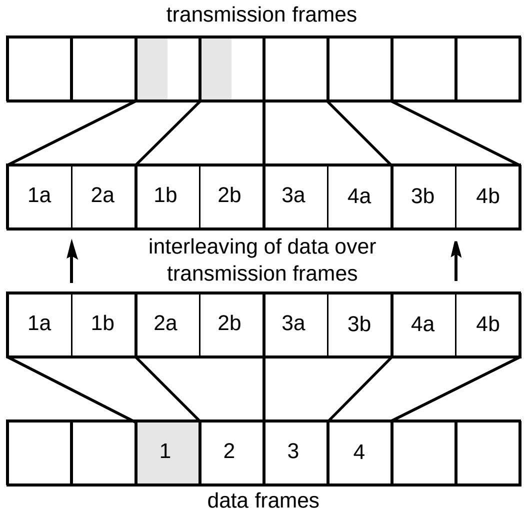

Fig 4 A simple example of interleaving. In this case the interleaving depth is 2.

To enable FEC to work effectively during long radio fades, a technique called frequency hopping can be employed. For a specific physical location, radio fades will only occur at particular carrier frequencies. By changing the frequency of the radio carrier at regular intervals, the effect of fading can be limited to short time periods, thus enabling interleaving and FEC to work more effectively. Frequency hopping benefits slow moving users most, for example a person walking along a street, because they experience the longest radio fades.

4.3 Error concealment

In poor radio conditions the channel decoder may be unable to correct all the bit errors in the speech data, and the speech decoder will enter an error concealment mode. Uncorrected bit errors may be detected either from parity bits added to the speech data, or in the case of a Viterbi decoder, from the soft decision value or the distance metric [10].

It has already been shown that many speech codecs use a parametric model to represent the speech to be encoded. Such codecs usually encode frames of speech, which are typically 10 to 30 ms in duration. In many source models, for example, the widely used linear predictive model, the parameter values change slowly over a number of frames. Speech codecs exploit this gradual variation of parameter values for error concealment purposes. When corrupted speech frames are detected, the speech decoder can use the parameter values from the last good frame to reconstruct an approximation of the missing speech. The same sound cannot be generated ad infinitum, and, after a small number of consecutive corrupted frames the output level of the speech decoder is gradually reduced until it has been completely muted.

5. Discontinuous transmission

The previous section introduced techniques for robust transmission over a radio channel. This section outlines a technique that is intended to increase the efficiency of the system— discontinuous transmission (DTX).

DTX is a mechanism that limits a user's radio transmissions to the periods when they are speaking. The benefits of DTX are twofold:

- it reduces the mean activity of the radio channel, reducing interference to distant users operating on the same carrier frequency,
- it lessens the power consumption of hand-held terminals, through reduced radio transmission and associated processing tasks.

The key component of a DTX system is the voice activity detector (VAD). The VAD identifies which frames contain speech and should therefore be transmitted. Occasionally, the VAD may fail to detect some of the speech, most commonly in high levels of acoustic noise. The resultant gaps in the transmitted speech can reduce its intelligibility, and are referred to as clipping. In non-stationary acoustic noise, the VAD may also incorrectly classify periods of noise as speech. This increases the proportion of frames that are transmitted and thus reduces the benefit of the DTX system. VAD design is therefore a compromise between reducing clipping and maintaining a useful DTX gain.

When the VAD indicates that speech is not present, noise is generated at the receiver to provide continuity between speech bursts. This is called comfort noise and is intended to simulate the ambient acoustic noise present in the original speech signal. Under very noisy conditions the insertion of comfort noise greatly improves speech intelligibility. To ensure good continuity in the reconstructed signal, the energy of the comfort noise is matched to that of the ambient noise at the transmitter; continuity can be further improved by matching the frequency spectrum. In some schemes, the noise model at the receiver is regularly updated at the expense of a marginal increase in radio activity.

6. An example digital mobile radio system — half-rate GSM

The preceding sections have introduced some of the techniques and considerations that go into the design of a digital mobile radio speech channel. This section describes a specific example in more detail - the half-rate GSM speech channel, which has recently been standardised by the European Telecommunications Standards Institute (ETSI). GSM is a good example of a system where the need to maximise of users, while maintaining acceptable speech quality, complexity and delay, has pushed speech coding technology to its limits.

The current GSM system uses a 13 kbit/s speech codec, which, after error correction coding, requires a 22.8 kbit/s traffic channel. To double the maximum number of users, a new set of recommendations has been defined which specify a 5.6 kbit/s speech codec and an 11.4 kbit/s channel. The two schemes are respectively referred to as full-rate and half-rate GSM [11 -- 13].

The half-rate GSM speech codec belongs to the code-excited linear prediction (CELP) class of codecs [14]. Before considering some of the features of the speech codec in detail, it is useful to review the fundamentals of speech production and CELP coding.

6.1 Speech production

The characteristics of a particular speech sound are determined by the shape of the vocal tract and the excitation mechanism. The vocal tract is considered to extend from the vocal cord to the lips and the nose. The shape of the vocal tract determines the gross frequency spectrum of a speech sound, although the effect of a sound radiating from the lips also modifies its frequency spectrum. The resonances introduced by the vocal tract are called formants, and in many languages their frequency carries information.

Three basic types of excitation are evident in speech production — voiced, unvoiced and plosive. Voiced speech is generated by the periodic opening and closing of the vocal cords. The frequency at which the cords vibrate is referred to as the pitch of the speech. Unvoiced speech is generated by forcing air past a constriction in the vocal tract and has the characteristics of noise. Plosives are generated by creating a closure in the vocal tract, increasing the air pressure behind it and then releasing the pressure suddenly. The / i / in seem, the / f / in face and the / p / in puff are examples of voiced, unvoiced and plosive sounds respectively. Further sounds can be generated from a mixture of excitation mechanisms, for example the /z/ in maze (mixed voiced and unvoiced).

6.2 Fundamentals of CELP coding

CELP coding is based on the linear predictive model of speech production [3], in which speech is generated by filtering an excitation signal. The filter coefficients are referred to as linear predictor (LP) coefficients, and are calculated by assuming that the excitation signal has a flat frequency spectrum. In practice, this is not always true, and the LP filter represents the combined frequency characteristics of the vocal tract, lip radiation and the excitation spectrum.

The linear predictive model of speech is reflected in the structure of the basic CELP decoder shown in Fig 5. The reconstructed speech signal s’ (t) is produced by filtering an excitation signal e’ (t). The synthesis filter uses LP coefficients calculated from the original speech. The excitation signal is constructed from the sum of two codewords, which are selected from the adaptive and fixed codebooks. The adaptive codebook contains delayed copies of the excitation signal e’ (t). This enables a periodic excitation signal to be generated during voiced speech. The fixed codebook contains random entries, which contribute mainly during unvoiced speech, plosives and transitions. The contribution from each code book is determined by its respective gain factor. The excitation information is typically updated more frequently than the LP coefficients.

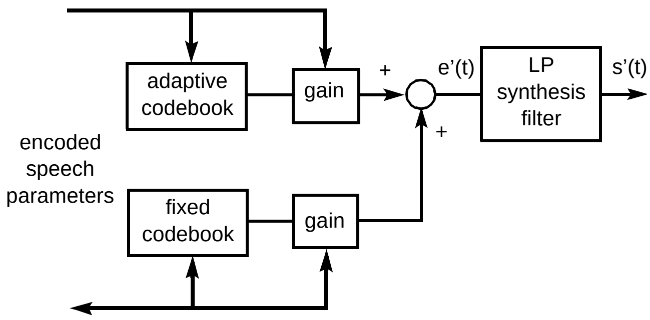

Fig 5 CELP decoder.

The CELP encoder is shown in Fig 6. The LP coefficients are calculated from the current frame of speech, quantised and transmitted to the decoder. The optimum codebook entries are determined by the analysis-by-synthesis method, where speech s’’(t) is synthesised from each codeword in turn. The index of the codeword that minimises the mean squared error between the synthesised speech and the input speech is transmitted to the decoder. The optimum codeword gain is also calculated and is transmitted after being quantised. The adaptive codebook search is performed first and its contribution incorporated into the fixed codebook search. It is searching the codebooks that makes CELP coding inherently complex.

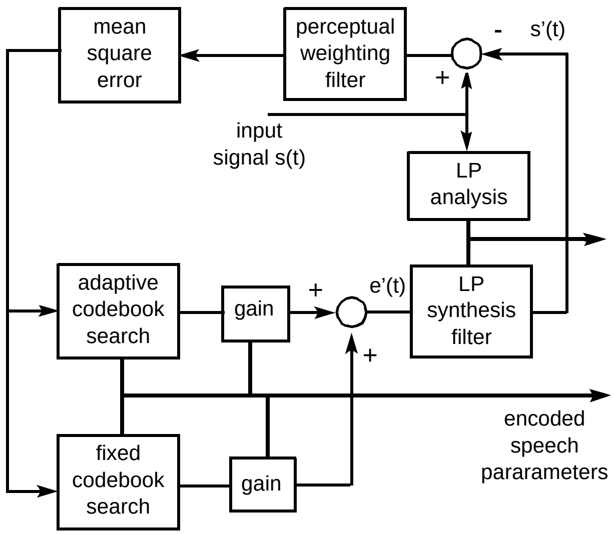

Fig 6 CELP encoder

The subjective performance of the CELP codec can be improved by weighting the error calculation in the codebook searches. This weighting favours codewords that minimise the distortion at frequencies where it will be most perceptible. The error signal is weighted by the perceptual weighting filter, whose coefficients are derived from the LP coefficients.

The structure of the CELP decoder determines which sounds can be faithfully reconstructed. It has been shown that the linear predictive model has much in common with the physical mechanisms of speech production. However, the CELP decoder is less capable of faithfully reconstructing many non-speech sounds. Consequently, the performance in the presence of ambient acoustic noise and under tandem conditions may be significantly worse than for a single coding of clean speech.

6.3 The half-rate GSM speech codec

Having introduced the basics of speech production and CELP coding, the design of the half-rate GSM speech codec will now be considered. For a complete description of the codec, the reader is referred to the ETSI Recommendation [15]. This section will outline some of the techniques the half-rate GSM speech codec uses to achieve speech quality at 5.6 kbit/s that is comparable to that of the 13-kbit/s full-rate speech codec.

The half-rate GSM speech codec is a vector sum excited linear prediction (VSELP) codec [16]. In a VSELP codec, the codewords in the fixed codebook are constructed by summing a set of basis vectors, which makes the codebook more robust to bit errors than a random codebook. In a random codebook, a single bit error in the index will return a codeword with no relation to the desired codeword. However, in the vector sum approach, a bit error in the index will only corrupt one of many contributions to the desired codeword. The vector sum technique also reduces the complexity of the codebook search.

The half-rate codec processes speech in 20 ms frames, which are divided into 5 ms sub-frames. The LP coefficients and frame energy are calculated and transmitted once every frame, while the codebook parameters are calculated and transmitted every sub-frame.

The characteristics of speech differ for different degrees of voicing. To enable good quality modelling of speech at a low bit rate, the half-rate codec has four modes of operation -- one unvoiced mode and three voiced modes. The mode is determined by the success of the adaptive codebook search. For each mode, separate quantisers are defined for the frame energy and codebook gains. While faithfully representing the four classes of speech, this enables the number of quantiser bits to be kept to a minimum. Since unvoiced speech is not periodic, the adaptive codebook contribution is replaced by a second VSELP codebook in the unvoiced mode.

Further reductions in bit rate are achieved by employing vector quantisation (VQ) to quantise the LP coefficient vector and the two codebook gains. The VQ process searches a codebook to find the codeword that best matches the vector to be quantised, and transmits the index of the selected codeword. VQ codebooks are optimised using training data that is representative of the data to be quantised [17]. For a given number of bits, VQ introduces less distortion than individually quantising each vector element. The disadvantage of VQ is the complexity of the codebook search. To reduce the complexity of the half-rate codec, the LP coefficients are quantised using a sub-optimal two-stage search.

To smooth the spectrum of the reconstructed speech, the LP coefficients may be interpolated over the sub-frames. However, this implies that the LP analysis window, which is used to calculate the LP coefficients, should be centred on the end of a frame, rather than its middle. In practice, the LP analysis window only extends 4.4 ms into the subsequent frame, increasing the algorithmic delay of the codec from 20 ms to 24.4 ms.

The half-rate speech decoder is a basic CELP decoder with the addition of a post-filter. The post-filter modifies the spectrum of the reconstructed speech signal to reduce the audibility of the codec distortion. This is achieved by emphasising the formant frequencies and, during voiced speech, the harmonic frequencies. The post-filter is a significant factor in improving the overall subjective performance of the codec.

6.4 Error correction coding

In half-rate GSM, the error correction coding increases the number of bits representing each 20 ms frame from 112 to 228 bits. The encoded speech bits are divided into class 1 (95 bits) and class 2 (17 bits) according to their impact on the subjective quality of the decoded speech. Three parity bits are generated for the most important class 1 bits. FEC is only applied to the class 1 bits and the parity bits. A further 6 tail bits must also be encoded to flush the memory of the convolutional encoder.

The ratio of the number of bits before error correction coding to the number of bits after coding is called the rate of a code. The class 1 bits (including parity and tail bits) are protected by a 104/211 rate-punctured convolutional code. The punctured code is derived by encoding the speech data with a 1/3 rate code and discarding some of the encoded bits. Punctured codes enable different portions of the data to be encoded at different rates [8]. In this case, 1/2 rate coding is applied to the class 1 bits and tail bits and 1/3 rate coding to the parity bits.

The number of errors that can be corrected by a convolutional code is determined by its constraint length.

Due to the lower interleaving depth of the half-rate speech channel, the convolutional code has a constraint length of 6 compared to 4 in the full-rate speech channel. This increase in constraint length raises the complexity of the Viterbi decoder by a factor of 4.

6.5 TDMA structure and interleaving

Each GSM radio channel supports 8 full-rate users or 16 half-rate users. Time division multiple access (TDMA) is used to share the radio channel capacity among the different users. Figure 7 shows the breakdown of a 120 ms TDMA multi-frame. The structure of a TDMA frame is divided into eight time slots. The physical contents of a time slot are called a transmission burst, and contain 114 data bits. In the half-rate case, each time slot is shared between two users, who use alternate bursts.

After error correction coding, the speech data is re-ordered, interleaved and mapped on to the transmission bursts. The encoded bits are re-ordered to remove any correlation between adjacent bit values. The 228 re-ordered bits are then divided into two groups and interleaved over the 114-bit transmission bursts.

The interleaving depths of the full- and half-rate speech channels are 8 and 4 respectively. This difference in interleaving depth makes the half-rate speech channel significantly less robust to error bursts than the full-rate channel.

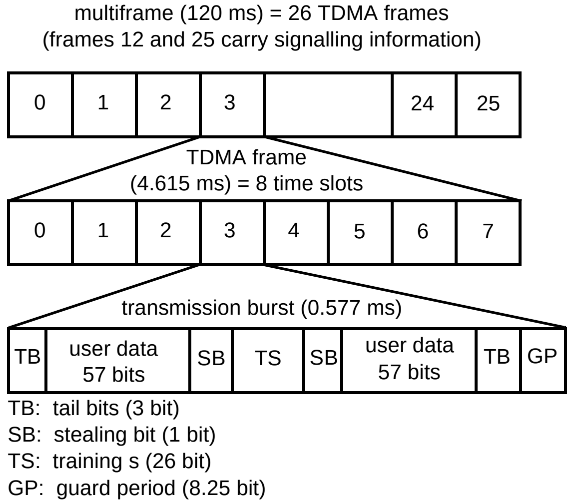

Fig 7 TDMA structure for GSM channel

6.6 Discontinuous transmission

The GSM recommendations define an optional DTX mode of operation for the speech traffic channel [18]. Figure 8 shows a block diagram of the full-rate GSM VAD structure [19]. The input signal frame is first filtered by an inverse linear predictive model of the acoustic noise. The energy of the filtered signal frame is then compared to an adaptive threshold to form an intermediate decision. A hangover mechanism is used to calculate the final VAD decision, which extends the duration of periods classified as speech. The noise model and adaptive threshold are updated if the secondary VAD indicates the absence of speech.This occurs if the signal is non-periodic and spectrally stationary. Hangover is added to the secondary VAD decision to reduce the probability of adaptation during speech.

At the end of a speech burst a further seven frames are transmitted. During this period, the acoustic noise is characterised at the encoder by averaging the spectral information and energy of the input signal. A silence descriptor (SID) frame, which contains the noise information, is then transmitted. The decoder uses the SID information to generate comfort noise between the speech bursts.

SID frames are transmitted every 240 ms to update the comfort noise parameters at the decoder. These parameters are interpolated over the SID update period to avoid discontinuities in the comfort noise.

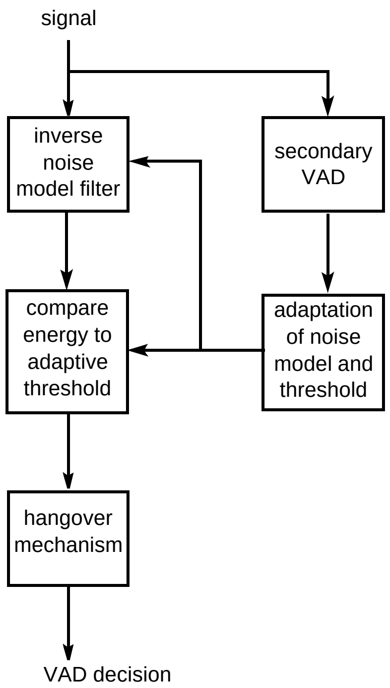

Fig 8 Structure of the GSM VAD.

6.7 Delay

The GSM recommendations define a round trip delay budget that must be met by manufacturers. Table 1 provides a coarse breakdown of the delay budget for the half-rate GSM channel.

Table 1 Breakdown of the half-rate GSM speech channel delay budget.

Delay	ms
Algorithmic delay	48.8
Interleaving delay	65.8
Implementation budget*	29.6
Other*	44.3
Total round trip delay*	188.5

* These figures are subject to review by ETSI.

The figures quoted in Table 1 are round-trip figures, which are the sum of the uplink and downlink delays. The implementation budget is the time allocated to perform the speech and channel coding tasks, and effectively determines the minimum processing power of the devices used if the delay requirement is to be met. The 'other' figure is the sum of the remaining system delays and budgets.

6.8 Complexity

The computational description of the half-rate system is based on the arithmetic operations defined for the full-rate system. The arithmetic is fixed-point and uses 16- and 32-bit operations. The exact complexity of an individual implementation will depend on the signal processing hardware used. However, for the purposes of complexity evaluation, a set of basic computational operations is defined, each operation being allocated a weight equal to the number of instructions required to perform it on a typical DSP device.

The theoretical worst-case complexity of the half-rate codec has been evaluated at 21.2 weighted million operations per second (wMOPS). This is 4.5 times the worst-case complexity of the full-rate codec. In practice, the computation required by the half-rate codec to process different signal frames may vary by up to 20%. These figures include channel coding, but not DTX functions.

7. Speech quality assessment

The wide range of competing factors to consider and balance in the design of a similar challenge in determining suitable processess to verify that the speech performance meets the specified criteria. In Common with past practice, the speech quality of the half-rate GSM system was measured by subjective assessment. However, objective assessment techniques, based on models of human perception, are also being developed [20]. In this section, some of the main issues relating to subjective speech quality assessment are outlined.

7.1 Methods of assessment

The simplest speech quality assessments are informal. They are quick and simple to perform and are commonly used, for example, to provide basic information on the effects of design changes during speech codec development. Such assessments often do not reflect real network scenarios and only provide limited information. As such, they are not sufficient to give a deep understanding of the performance issues that are important to the users of a system and to service providers.

Formal assessment methodologies have been developed to address such requirements, with much greater emphasis placed on controlling the process. Some of the factors considered include:

- acoustic environment,
- terminal equipment,
- the order in which impairments (such as channel errors or tandeming) are introduced,
- the selection of participants for the test,
- the questions asked of the participants.

International collaborative work has led to the production of the ITU-T P-series of recommendations, which form the basis for many of these tests.

Formal assessment methods are in a continual state of flux. This is caused by the development of new systems and services, which attempt to meet the increasing demand for improved and more accessible communications. Many of these developments affect the way in which speech performance is assessed, forcing existing methods to evolve and new methods to be developed. For example, the use of mobile radio systems has introduced impairments caused by radio channel errors and a wide range of different acoustic environments such as train stations, the interiors of moving vehicles and busy streets. These were not considerations in the development of the fixed telephone network, and traditional methodologies have been forced to evolve, in order to provide relevant information. This process of evolution has led to the development of a wide range of formal methods [21] each designed to address a specific issue.

7.2 Selecting appropriate methods

Almost all modern speech quality assessments use opinion-based methods, where the participants are asked directly for their opinion on some aspect of the quality of the system they are assessing. To determine the most suitable methodology, different aspects of the system being investigated have to be considered, e.g. the types of impairment that are likely, how and where the system is to be used, and how it will interwork with other telecommunications networks. These factors will all have some effect on the quality of received speech, the ability to speak without difficulty and the ability to converse naturally. It would be an impossible task to cover all of these aspects in any single performance assessment, since the cost and time-scale would be prohibitive. Hence, a balance has to be reached through understanding the relative importance of different factors, the relevance of existing information, and practical considerations such as time and cost.

In the GSM standardisation process, there were several different phases of assessment. The majority of these phases were used to select the most suitable speech codec, based on its performance for the more important impairments. Once a codec had been chosen, the emphasis changed to characterising its performance over a wider range of operating condition. This was to provide more detailed information for potential service providers.

In the half-rate GSM standardisation process, the emphasis in assessment was placed on the quality of received speech, with the aim of attaining a system with a performance level comparable to that of the existing full-rate GSM system. This enabled the exclusive use of listening-only tests, where there is no need for the participants to converse with one another.

Experience at BT Laboratories and other laboratories around the world has led to the use of two primary types of listening-only methodologies -- those based on listening quality and those based on listening effort. In listening quality assessments, participants listen to speech and then give their opinion as to whether the quality was excellent, good, fair, poor, or bad. In listening effort assessments, the participants are asked how much effort is required to understand the meanings of the sentences they have just heard. The listening effort method provides more discrimination at the lower end of acceptable performance for telephony and hence is typically used to investigate performance for the more critical impairments, whereas the listening quality method provides valuable information over typical operational conditions. Both of these methods were used in the half-rate GSM performance assessment process.

7.3 Drawing conclusions

Once an assessment has been completed, the results are analysed using statistical methods to determine what conclusions can be drawn and how best to express them. In the GSM process, the results of this are summarised by Usai et al [22] and in an ETSI Technical Report [23]. The latter serves as an informative document, which supports the recommendations with background information on the performance of the half-rate system. The main conclusion from this information was that the performance of the half-rate speech codec is comparable to that of the full-rate speech codec, except when used in mobile-to-mobile situations and in certain noisy environments.

8. Future directions

It has been shown how the performance requirements of a system depend heavily upon its application. As the mobile telephony market has grown, the proportion of non-business users has increased dramatically. A typical user may no longer be a business person willing to trade quality of service for the benefits of mobility, but a member of the general public who will expect a quality of service similar to that provided by the PSTN. Consider the difference between instructing a broker to sell shares and whispering sweet nothings to a loved one. It is arguable that the latter requires a degree of naturalness not offered by current mobile telephony systems.

It has already been shown that the performance of speech codecs is often biased towards a particular gender, age or language group. As the market-place for mobile communications grows to encompass a greater proportion of the public and a greater portion of the world, this trend may become increasingly unacceptable. Does the concept of an egalitarian society extend to the performance of mobile communications systems? The authors would argue that it probably does.

The mobile communications systems of the future will need to provide communication that is, at the very least, as natural as the current PSTN. This means lower distortion, lower delay, and possibly wider bandwidth. These considerations appear to indicate that a higher data rate must be offered per user. However, in direct conflict with this is the requirement to handle an increasing volume of traffic. For the moment at least, the public's desire for mobile communications would appear to be insatiable, and speech quality improvements will have to be made within existing, and possibly lower, data rates.

The adoption of a small number of global codec standards could reduce the need for unnecessary speech coding stages. However, recent advances in speech coding and device technology have resulted in a proliferation of speech coding standards. With many new markets for speech transmission emerging, perhaps the communications industry should heed the following ironic observation:

'The nice thing about standards is that there are so many of them to choose from.'

A. S. Tanenbaum

On a final note, the use of more immersive environments for quality assessment must be considered in the future, so the effects of more than just the acoustic surroundings can be understood. This will be particularly relevant to the design and assessment of mobile radio systems. For example, the effects of the radio shadows created by buildings could be examined by giving participants the opportunity to move around in a virtual environment. Such techniques will enable controlled studies of system performance in more realistic contexts.

9. Conclusions

As the market for mobile communications grows and evolves, customers are increasingly likely to demand a better quality of service from both manufacturers and operators. The designers of speech and channel codecs are already facing many of the fundamental problems presented by the mobile telephony markets of the future. One of the main challengies is to design speech channels that offer near-transparent quality, low delay, and consistant performance over a wide range of error conditions. In conflict with this are the physical, engineering and economic factors that limit the bit rate available to indevidual users. One of the immediate challenges is to design sub-4-kbit/s speech codecs that are speaker- and language-independent, and, perhaps most difficult of all, virtually unaffected by the presence of ambient acoustic noise and tandem conditions.

In this paper the authors have described the major considerations involved in speech transmission over digital mobile radio channels. Both the design and assessment of such systems present considerable engineering challenges, requiring a careful trade-off between many competing factors. An appropriate balance is often very difficult to achieve, and must evolve with changing customer expectations. However, achieving this balance is vital to customer satisfaction.

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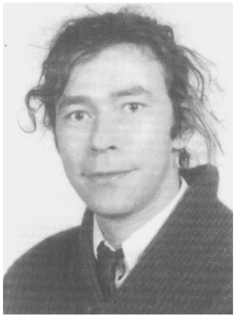

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